MATH423: Introduction to Computational Topology

Worked Calculations for Sections 6.1–6.4

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These notes contain fully worked calculations for Sections 6.1–6.4 of the lecture notes: chains and the boundary operator over $\mathbb{Z}/2$ (6.1), cycles, boundaries and homology (6.2), the zeroth homology group (6.3), and the Euler–Poincaré formula (6.4). All coefficients are in $\mathbb{Z}/2$, so $1+1=0$, and b_k denotes $\dim H_k(K;\mathbb{Z}/2)$. Recall the two formulas we use throughout:

$$b_k = n_k - \operatorname{rank}\partial_k - \operatorname{rank}\partial_{k+1}, \qquad \chi(K) = \sum_k (-1)^k n_k = \sum_k (-1)^k b_k,$$

where n_k is the number of k-simplices of K.

How to compute the rank of a matrix over $\mathbb{Z}/2$

Row reduction works exactly as over $\mathbb{R}$, with two simplifications: the only nonzero scalar is 1, so there is never any need to scale a row, and subtracting a row is the same as adding it. The only operations needed are swapping two rows and adding one row to another. Adding two rows means adding entries with $1+1=0$, that is, taking the exclusive or of the rows. The rank is the number of nonzero rows in an echelon form.

1 Chains and the Boundary Operator (6.1)

Worked Example 1.1 (Adding chains)**.** In the boundary of the tetrahedron on $\{1,2,3,4\}$, let

$$c_1 = \{1,2\}+\{2,3\}+\{1,3\}, \qquad c_2 = \{1,3\}+\{3,4\}+\{1,4\}.$$

Then

$$c_1 + c_2 = \{1,2\}+\{2,3\}+\{3,4\}+\{1,4\}+2\{1,3\} = \{1,2\}+\{2,3\}+\{3,4\}+\{1,4\},$$

since $2\{1,3\}=0$. As sets of edges, c_1+c_2 is the symmetric difference of c_1 and c_2: the common edge $\{1,3\}$ is removed. Geometrically, two triangle loops sharing an edge add up to the loop around the square $1,2,3,4$.

Worked Example 1.2 (Boundaries of chains)**.** In the boundary of the tetrahedron, compute the boundary of the 2-chain consisting of the three triangles containing the vertex 1:

$$d = \{1,2,3\} + \{1,3,4\} + \{1,2,4\}.$$

Applying ∂_2 to each triangle and adding,

$$\begin{aligned}\partial_2 d &= (\{2,3\} + \{1,3\} + \{1,2\}) + (\{3,4\} + \{1,4\} + \{1,3\}) + (\{2,4\} + \{1,4\} + \{1,2\}) \\ &= \{2,3\} + \{3,4\} + \{2,4\} + 2\{1,2\} + 2\{1,3\} + 2\{1,4\} \\ &= \{2,3\} + \{3,4\} + \{2,4\}.\end{aligned}$$

The three edges through the vertex 1 each lie in two of the triangles and cancel. What remains is the loop around the fourth triangle $\{2,3,4\}$: the boundary of a disk made of three triangles is its outer circle. Applying ∂_1 once more,

$$\partial_1 \partial_2 d = (\{2\} + \{3\}) + (\{3\} + \{4\}) + (\{2\} + \{4\}) = 0,$$

as Lemma 6.5 guarantees.

Worked Example 1.3 (Boundary matrices of the sphere)**.** Let K be the boundary of the tetrahedron. We order the simplices as follows:

vertices: $1,2,3,4$; edges: $12,13,14,23,24,34$; triangles: $123,124,134,234$,

writing 12 for $\{1,2\}$ and so on. The column of a simplex in a boundary matrix has a 1 in the rows of its facets:

$$\partial_1 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}\!\!\begin{array}{c} \begin{array}{cccccc} 12 & 13 & 14 & 23 & 24 & 34 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} \end{array}, \qquad \partial_2 = \begin{array}{c} \\ 12 \\ 13 \\ 14 \\ 23 \\ 24 \\ 34 \end{array}\!\!\begin{array}{c} \begin{array}{cccc} 123 & 124 & 134 & 234 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{array}.$$

Each column of ∂_1 has two ones and each column of ∂_2 has three. We check $\partial_1 \partial_2 = 0$ on the first column: the column of 123 in ∂_2 selects the columns 12, 13, 23 of ∂_1, and

$$\begin{bmatrix}1\\1\\0\\0\end{bmatrix} + \begin{bmatrix}1\\0\\1\\0\end{bmatrix} + \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}2\\2\\2\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\\0\end{bmatrix}.$$

The other three columns are checked in the same way, so $\partial_1 \partial_2$ is the zero 4×4 matrix.

2 Cycles, Boundaries and Homology (6.2)

Worked Example 2.1 (The sphere, step by step)**.** We compute the Betti numbers of the boundary of the tetrahedron from the matrices of Worked Example 1.3.

Rank of ∂_1. Row reduce over $\mathbb{Z}/2$:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_1+R_2\to R_2} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{R_2+R_3\to R_3} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3+R_4\to R_4} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are three nonzero rows, so $\operatorname{rank}\partial_1 = 3$. (This agrees with Theorem 6.12: the 1-skeleton is connected, so $\operatorname{rank}\partial_1 = n_0 - 1$.)

Rank of ∂_2. It is easier to look at the columns. The sum of the first three columns of ∂_2 is

$$(1+1+0,\ 1+0+1,\ 0+1+1,\ 1+0+0,\ 0+1+0,\ 0+0+1)^T = (0,0,0,1,1,1)^T,$$

which is the fourth column. So the four columns satisfy one relation, $123 + 124 + 134 + 234 \in \ker\partial_2$. The first three columns are independent: the rows 14, 13, 12 show that a combination $a\cdot 123 + b\cdot 124 + c\cdot 134$ with zero boundary must have $b+c=0$, $a+c=0$, $a+b=0$, and then row 23 gives $a=0$, so $b=c=0$. Hence $\operatorname{rank}\partial_2 = 3$ and $\ker\partial_2$ is spanned by the sum of all four triangles.

Betti numbers. With $n=(4,6,4)$ and $\partial_3 = 0$:

$$b_0 = 4-0-3 = 1, \qquad b_1 = 6-3-3 = 0, \qquad b_2 = 4-3-0 = 1.$$

Interpretation. Z_1 has dimension $6-3=3$, and it is spanned by the boundaries of the three triangles 123, 124, 134. So every 1-cycle is a boundary: $Z_1 = B_1$, and there is no 1-dimensional hole. On the other hand Z_2 is spanned by the sum of all four triangles, and $B_2 = 0$ because there are no tetrahedra in K. This 2-cycle is the one void of the sphere.

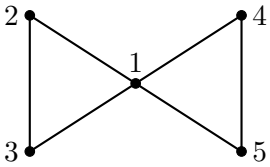
Worked Example 2.2 (Filling the sphere)**.** Let Δ^3 be the full tetrahedron, that is, K together with the 3-simplex 1234. Then

$$\partial_3(1234) = 234 + 134 + 124 + 123,$$

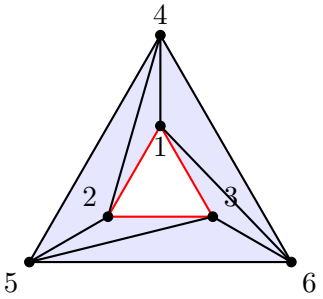
so ∂_3 is a 4×1 matrix with all entries equal to 1, of rank 1. Its image is exactly the 2-cycle found above, which is now a boundary. The ranks of ∂_1 and ∂_2 are unchanged, and $n=(4,6,4,1)$, so

$$b_0 = 1, \qquad b_1 = 6-3-3 = 0, \qquad b_2 = 4-3-1 = 0, \qquad b_3 = 1-1-0 = 0.$$

Filling in the void kills H_2, and the full tetrahedron has the homology of a point. Check: $\chi = 4-6+4-1 = 1 = b_0$.



(a) two hollow triangles at a vertex



(b) a triangulated annulus

Figure 1: (a) The complex of Worked Example 2.3. (b) The annulus of Worked Example 2.4; the inner loop (red) generates H_1.

Worked Example 2.3 (Two hollow triangles sharing a vertex)**.** Let K consist of the vertices $1, \dots, 5$ and the edges $12, 13, 23, 14, 15, 45$ (Figure 1(a)). There are no triangles. The boundary matrix is

$$\partial_1 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{cccccc} 12 & 13 & 23 & 14 & 15 & 45 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array}.$$

The complex is connected, so $\operatorname{rank} \partial_1 = 5 - 1 = 4$ by Theorem 6.12. (Directly: the rank is at most 4, since each column has two ones and so the five rows add up to zero; and the four columns 12, 13, 14, 15 are independent, since their second ones lie in the four different rows 2, 3, 4, 5.) Since $\partial_2 = 0$,

$$b_0 = 5 - 4 = 1, \qquad b_1 = 6 - 4 - 0 = 2.$$

Two independent 1-cycles are $z_1 = 12 + 23 + 13$ and $z_2 = 14 + 45 + 15$. They are independent in H_1 because $B_1 = 0$ (there are no triangles), so two cycles are homologous only if they are equal. The cycle $z_1 + z_2$, which runs around both loops like a figure eight, is a third cycle, but it is not new: it is the sum of the other two.

Worked Example 2.4 (The annulus)**.** Let K have the six triangles

$$124, \quad 245, \quad 235, \quad 356, \quad 136, \quad 146$$

(Figure 1(b)). Its 12 edges are the 6 interior edges $14, 24, 25, 35, 36, 16$ and the 6 boundary edges $12, 23, 13$ (inner loop) and $45, 56, 46$ (outer loop). So $n = (6, 12, 6)$ and $\chi = 0$.

Rank of ∂_1. K is connected, so $\operatorname{rank} \partial_1 = 5$.

Rank of ∂_2. We show $\ker \partial_2 = 0$. Let S be a set of triangles with $\partial_2 S = 0$. Every boundary edge lies in exactly one triangle, so no triangle containing a boundary edge can be in S (its boundary edge would appear once). But every one of the six triangles contains a boundary edge (for instance $124 \ni 12$ and $245 \ni 45$). So $S = \emptyset$, and $\operatorname{rank} \partial_2 = 6 - 0 = 6$.

Betti numbers.

$$b_0 = 6 - 0 - 5 = 1, \qquad b_1 = 12 - 5 - 6 = 1, \qquad b_2 = 6 - 6 - 0 = 0.$$

Check: $1 - 1 + 0 = 0 = \chi$.

A generator, and homologous cycles. The inner loop $z_{\text{in}} = 12 + 23 + 13$ is a cycle. The outer loop $z_{\text{out}} = 45 + 56 + 46$ is also a cycle. Adding up all six triangles,

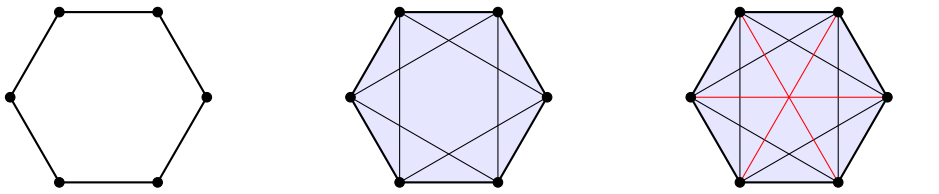
$$\partial_2(124 + 245 + 235 + 356 + 136 + 146) = z_{\text{in}} + z_{\text{out}},$$

because every interior edge lies in two of the triangles and cancels, while every boundary edge lies in one. Therefore $z_{\text{out}} = z_{\text{in}} + \partial_2(\dots)$: the inner and outer loops are homologous, and they represent the same nonzero class in H_1. The class is nonzero, that is, z_{in} is not a boundary. Suppose $\partial_2 S = z_{\text{in}}$ for a set S of triangles. The outer edges 45, 56, 46 do not occur in z_{in}, and each lies in only one triangle, so $245, 356, 146 \notin S$. The interior edge 24 lies only in 124 and 245; since $245 \notin S$, also $124 \notin S$, otherwise 24 would occur once in $\partial_2 S$. In the same way 25 excludes 235, and 36 excludes 136. So $S = \emptyset$ and $\partial_2 S = 0 \neq z_{\text{in}}$, a contradiction. Hence H_1 is spanned by the class of z_{in}, which is also the class of z_{out}.

Worked Example 2.5 (Is a given chain a cycle? A boundary?)**.** Let K be the square $1, 2, 3, 4$ subdivided by the diagonal 13: triangles 123 and 134, edges $12, 23, 13, 34, 14$.

- $c = 12 + 23 + 34 + 14$: $\partial_1 c = (1 + 2) + (2 + 3) + (3 + 4) + (1 + 4) = 0$, so c is a cycle. It is also a boundary: $\partial_2(123 + 134) = 12 + 23 + 13 + 13 + 34 + 14 = c$.
- $c' = 12 + 23 + 13$: a cycle, and the boundary of 123.
- $c'' = 12 + 23$: $\partial_1 c'' = 1 + 3 \neq 0$, so c'' is not a cycle, and in particular not a boundary.

Here $n = (4, 5, 2)$, $\operatorname{rank} \partial_1 = 3$, $\operatorname{rank} \partial_2 = 2$, and $b = (1, 0, 0)$: every cycle is a boundary, as it must be for a disk.



$r = 1$: a hexagon $\quad$ $r = \sqrt{3}$: an octahedron $\quad$ $r = 2$: the full simplex

Figure 2: The Vietoris–Rips complex of the six vertices of a regular hexagon with side 1, where two points are joined when their distance is at most r. At $r = \sqrt{3}$ every point is joined to all others except its opposite point; at $r = 2$ the three long diagonals (red) appear too.

Worked Example 2.6 (A Rips complex with a surprise)**.** Let X be the six vertices of a regular hexagon with side 1. The distances between them are 1 (neighbors), $\sqrt{3}$ (second neighbors) and 2 (opposite vertices). Let R_r be the Rips complex in which a set of points is a simplex when all pairwise distances are at most r (Figure 2).

- For $r < 1$: six isolated points, $b_0 = 6$.
- For $1 \leq r < \sqrt{3}$: only neighbors are joined, and there are no triangles (three consecutive vertices have pairwise distances $1, 1, \sqrt{3}$). The complex is a hexagon: $n = (6, 6)$, $b_0 = 1$, $b_1 = 6 - 5 = 1$.

- For $\sqrt{3} \le r < 2$: every point is joined to every other point except the opposite one. A set of points is a simplex if and only if it contains no pair of opposite points. Choosing one point from each of the three opposite pairs gives the $2^3 = 8$ triangles, and there are no 4-element simplices, since four points must contain an opposite pair. So $n = (6, 12, 8)$. This is exactly the boundary of the octahedron, with the three opposite pairs as the poles. Rather than reducing matrices, we use what we know: it is a triangulated sphere, a closed connected surface, so $b_0 = b_2 = 1$ by Proposition 6.14, and the Euler–Poincaré formula gives

  $$b_1 = b_0 + b_2 - \chi = 1 + 1 - (6 - 12 + 8) = 0.$$

  The hexagon's hole has been filled, but a 2-dimensional void has appeared!

- For $r \ge 2$: all points are joined, the complex is the full 5-simplex, and it has the homology of a point.

The appearance of H_2 for a planar point set is a feature of Rips complexes, which are not subsets of the plane. The Čech complex, which is homotopy equivalent to a union of disks in the plane, can never have $b_2 \neq 0$ for points in $\mathbb{R}^2$. In persistent homology (Weeks 7–8) such short-lived classes are typically regarded as noise.

3 H_0 Counts Connected Components (6.3)

Worked Example 3.1 (A disconnected complex)**.** Let K have the maximal simplices 123, 45 and the isolated vertex 6. So $n = (6, 4, 1)$: vertices $1, \dots, 6$; edges $12, 13, 23, 45$; triangle 123.

$$\partial_1 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccc} 12 & 13 & 23 & 45 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array}.$$

The first three columns have rank 2 (their sum is zero, and any two are independent), and the last column is independent of them, so $\operatorname{rank} \partial_1 = 3 = n_0 - c$ with $c = 3$ components. The triangle has boundary $12 + 13 + 23 \neq 0$, so $\operatorname{rank} \partial_2 = 1$. Hence

$$b_0 = 6 - 3 = 3, \qquad b_1 = 4 - 3 - 1 = 0, \qquad b_2 = 1 - 1 = 0.$$

A basis of H_0 is given by one vertex from each component, say $\{1\}$, $\{4\}$, $\{6\}$. Every other vertex is homologous to one of these: for example $\{3\} = \{1\} + \partial_1(13)$, and $\{5\} = \{4\} + \partial_1(45)$. The *parity map* from the proof of Theorem 6.12 sends a 0-chain to the vector of parities of its numbers of vertices in the three components; for example $\{1\} + \{2\} + \{4\} \mapsto (0, 1, 0)$, which shows that this 0-chain is homologous to $\{4\}$ and is not a boundary.

Worked Example 3.2 (b_0 of a Rips complex)**.** Let $X = \{0, 1, 3, 7, 8\} \subset \mathbb{R}$, and let R_r be the Rips complex at distance threshold r. The consecutive gaps are $1, 2, 4, 1$. By

Theorem 6.12, $b_0(R_r)$ is the number of connected components of the graph of pairs at distance $\le r$, and on the line these are the clusters separated by gaps larger than r:

$$b_0(R_r) = \begin{cases} 5 & 0 \le r < 1, \\ 3 & 1 \le r < 2 \quad (\{0,1\},\{3\},\{7,8\}), \\ 2 & 2 \le r < 4 \quad (\{0,1,3\},\{7,8\}), \\ 1 & r \ge 4. \end{cases}$$

The merge values $1, 1, 2, 4$ are the edge lengths of the minimum spanning tree of X, as in Programming Exercise 4 of Section 4. No linear algebra is needed: b_0 can always be computed by a graph search or union-find.

4 The Euler–Poincaré Formula (6.4)

Worked Example 4.1 (Checking the formula)**.** The following table collects the examples above. In each row, the two alternating sums agree.

Complex	$(n_0, n_1, n_2, \dots)$	$(b_0, b_1, b_2, \dots)$	χ
boundary of tetrahedron	$(4, 6, 4)$	$(1, 0, 1)$	2
full tetrahedron	$(4, 6, 4, 1)$	$(1, 0, 0, 0)$	1
two triangles at a vertex	$(5, 6)$	$(1, 2)$	-1
annulus	$(6, 12, 6)$	$(1, 1, 0)$	0
square with diagonal	$(4, 5, 2)$	$(1, 0, 0)$	1
hexagon, $1 \le r < \sqrt{3}$	$(6, 6)$	$(1, 1)$	0
octahedron, $\sqrt{3} \le r < 2$	$(6, 12, 8)$	$(1, 0, 1)$	2
disconnected example	$(6, 4, 1)$	$(3, 0, 0)$	3

Worked Example 4.2 (Using the formula to avoid computing ranks)**.** Let K be the 2-skeleton of the 4-simplex: all vertices, edges and triangles on the vertex set $\{1, 2, 3, 4, 5\}$, so $n = (5, 10, 10)$ and $\chi = 5 - 10 + 10 = 5$. We find the Betti numbers with very little computation.

- $b_0 = 1$, since the complex is connected.
- $b_1 = 0$. Let z be a 1-cycle. For every edge ab with $1 \notin \{a, b\}$, the triangle $1ab$ is in K, and $ab = \partial_2(1ab) + 1a + 1b$. Replacing each such edge in z in this way changes z by a boundary, so z is homologous to a cycle z' using only edges $1a$ through the vertex 1. But $\partial_1(1a) = 1 + a$, and the vertex a appears in no other edge through 1, so a sum of such edges can have zero boundary only if it is empty. Hence $z' = 0$, and z is a boundary.
- Now the Euler–Poincaré formula gives $b_2 = \chi - b_0 + b_1 = 5 - 1 + 0 = 4$.

So K has four independent 2-dimensional voids. A basis of H_2 is given by the boundaries of the four tetrahedra 1234, 1235, 1245, 1345, each the sum of its four triangles. Each is a 2-cycle, and they are independent, since each contains a triangle not containing 1 (234,

235, 245, 345, respectively) that occurs in none of the others. Since K has no 3-simplices, $B_2 = 0$, so these cycles are independent in H_2. Computing $\operatorname{rank} \partial_2 = n_2 - b_2 = 6$ directly would have required reducing a 10×10 matrix.

Worked Example 4.3 (Surfaces)**.** For a triangulated closed connected surface, Proposition 6.14 gives $b_0 = b_2 = 1$, so the Euler–Poincaré formula gives $b_1 = 2 - \chi$.

- The 9-vertex torus of Week 2: $n = (9, 27, 18)$, $\chi = 0$, so $b = (1, 2, 1)$.
- The 6-vertex projective plane of Worksheet 1: $n = (6, 15, 10)$, $\chi = 1$, so $b = (1, 1, 1)$.
- An orientable surface of genus 3: $\chi = -4$, so $b_1 = 6$.
- A closed surface triangulated with $V = 8$ vertices and $F = 16$ triangles: $E = 3F/2 = 24$, $\chi = 0$, so $b = (1, 2, 1)$. It is a torus or a Klein bottle; over $\mathbb{Z}/2$ the Betti numbers cannot tell which.

Worked Example 4.4 (Detecting a mistake)**.** A student reports that a connected 2-dimensional complex with $n = (7, 15, 9)$ has Betti numbers $(1, 2, 0)$. This cannot be right: $\chi = 7 - 15 + 9 = 1$, while $1 - 2 + 0 = -1$. If $b_2 = 0$ is correct, the formula forces $b_1 = b_0 + b_2 - \chi = 0$. The Euler–Poincaré formula is a cheap and very effective check on any homology computation.

Remark 4.5. In all the examples, the hardest part is the rank of ∂_k for $k \geq 2$. Three tricks avoided it: (i) $\operatorname{rank} \partial_1 = n_0 - b_0$ by Theorem 6.12; (ii) for closed connected surfaces, $b_2 = 1$ by Proposition 6.14; (iii) once all but one Betti number is known, the Euler–Poincaré formula gives the last one.